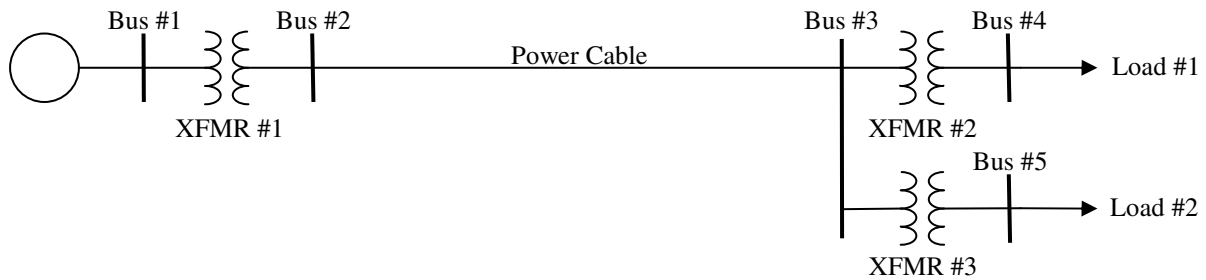


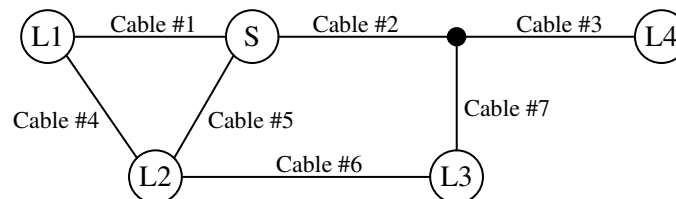
Problem #1

In the 3 ϕ power system shown in the one line diagram below, a source supplies power to two loads using three transformers and a length of underground power cable. Draw a **block diagram** and the **equivalent circuit** for this 3 ϕ power system. Use the pi model for the power cable in the equivalent circuit



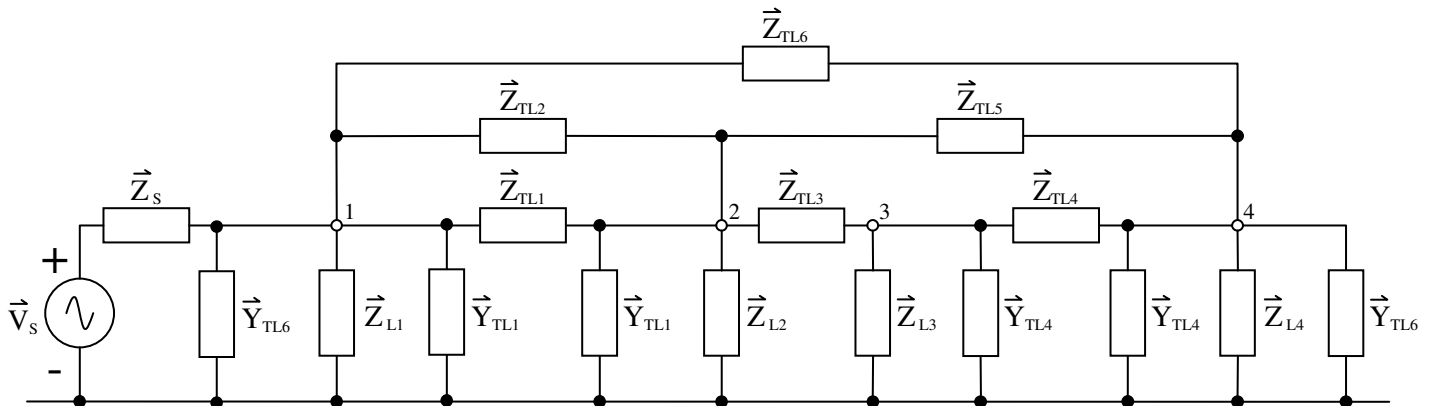
Problem #2

The block diagram for a 1 ϕ power system is given below. Draw a **one line diagram** and the **equivalent circuit** for this 1 ϕ power system. Use the series impedance model for the power cables in the equivalent circuit.



Problem #3

The equivalent circuit shown below models a 3 ϕ power system. Draw a **one line diagram** for this system.



Problem #4

A 3 ϕ Δ -connected source supplies power to a Δ -connected constant impedance load through an 800 foot length of power cable. The type of power cable used has three conductors.

Draw and completely label all four of the different types of power system diagrams (block diagram, one line diagram, wiring diagram, and equivalent circuit) for this 3 ϕ power system. Use the pi model for the power cable in the equivalent circuit.

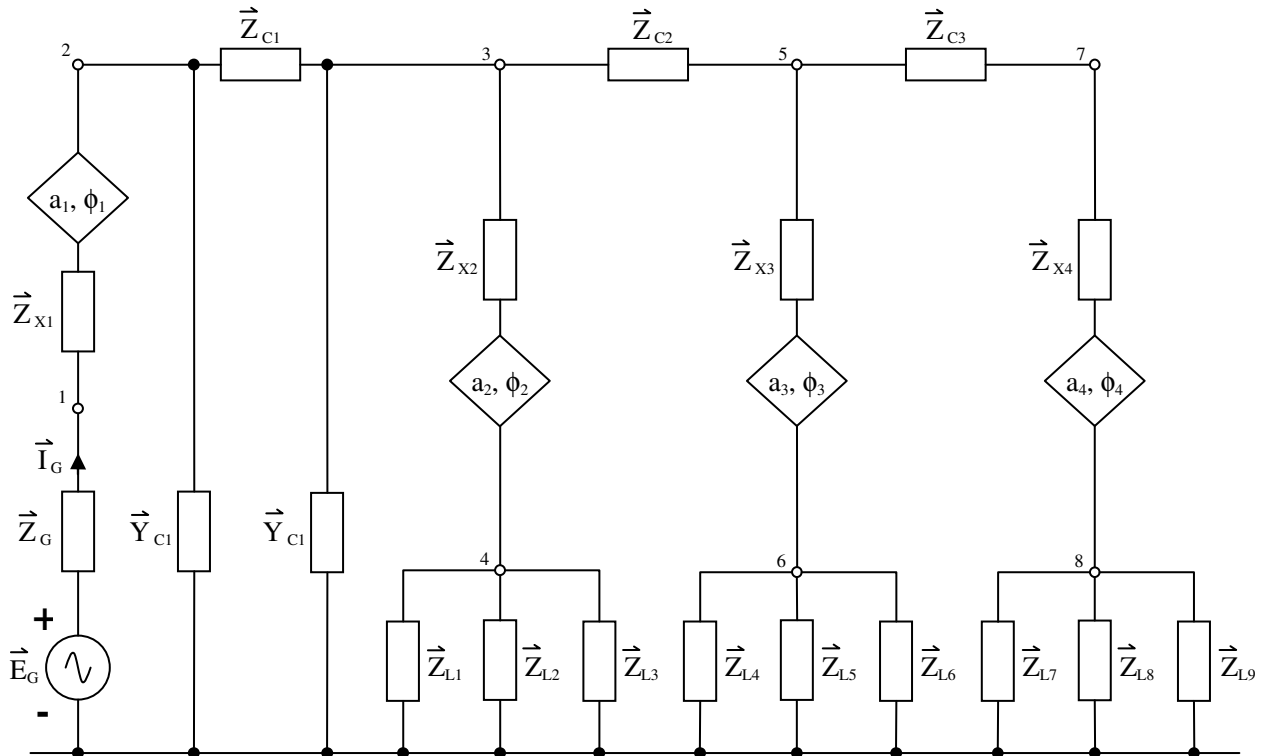
Problem #5

A 1 ϕ power system for a small manufacturing facility is supplied by a 240V tie from the electric utility company. Inside the manufacturing facility, there are two power distribution panels to which all of the building's loads are connected. The first distribution panel is connected to the electric utility company tie through a 500ft power cable, and a 350ft power cable connects the two distribution panels together.

Draw and completely label all four of the different types of power system diagrams (block diagram, one line diagram, wiring diagram, and equivalent circuit) for this 1 ϕ power system. Use the series impedance model for the power cables in the equivalent circuit.

Problem #6

The equivalent circuit shown below models a 3 ϕ power system. Draw a **one line diagram** for this system.



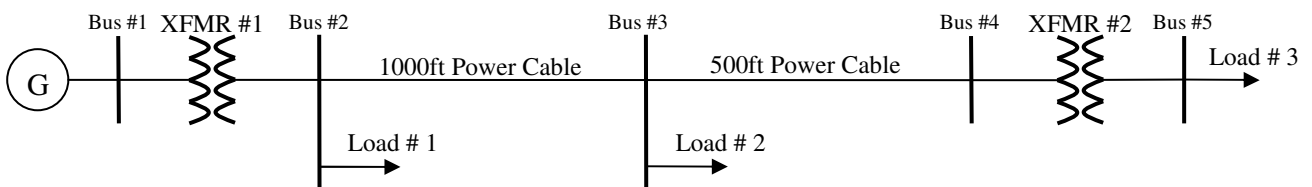
Problem #7

In a 1 ϕ power system, two remotely located high voltage sources connect through separate high voltage transmission lines to a centrally located bus. At this centrally located bus, a step-down transformer is used to supply power at a lower voltage level to two loads. Both of the sources are electric utility company ties.

Draw and completely label all four of the different types of power system diagrams (block diagram, one line diagram, wiring diagram, and equivalent circuit) for this 1 ϕ power system. Use the pi model for the transmission lines in the equivalent circuit.

Problem #8

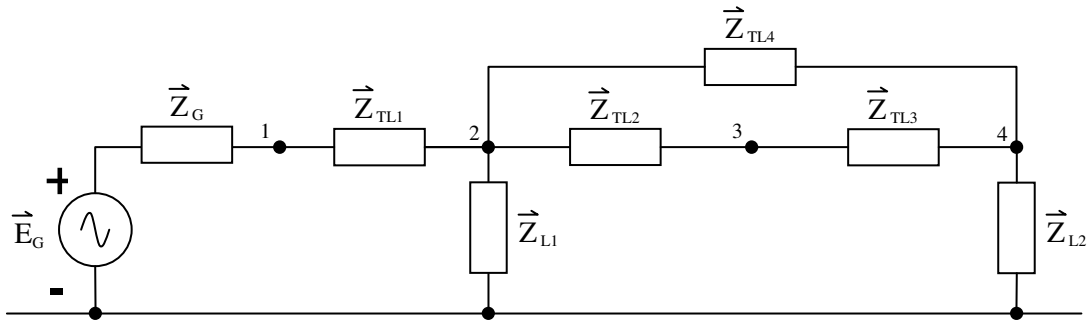
The one diagram for a 3 ϕ power system is given below.



Load #1 and Load #2 are both Δ -connected and Load #3 is Y-connected. The power cables used have 4-conductors. **Draw and completely label the other three different types of power system diagrams** for this 3 ϕ power system. Use the pi model for the 1000ft power cable and the series impedance model for the 500ft power cable in the equivalent circuit.

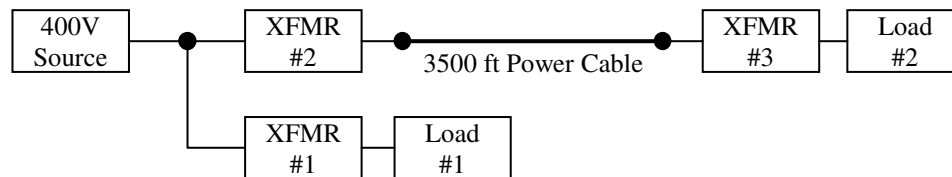
Problem #9

An equivalent circuit that models a 1 ϕ power system is drawn below. **Draw and completely label the other three different types of power system diagrams** (block diagram, one line diagram, and wiring diagram) for this 1 ϕ power system.



Problem #10

A 1 ϕ power system is shown in the block diagram below. Draw a **one line diagram** and the **equivalent circuit** for this 1 ϕ power system. Use the pi model for the power cable in the equivalent circuit.



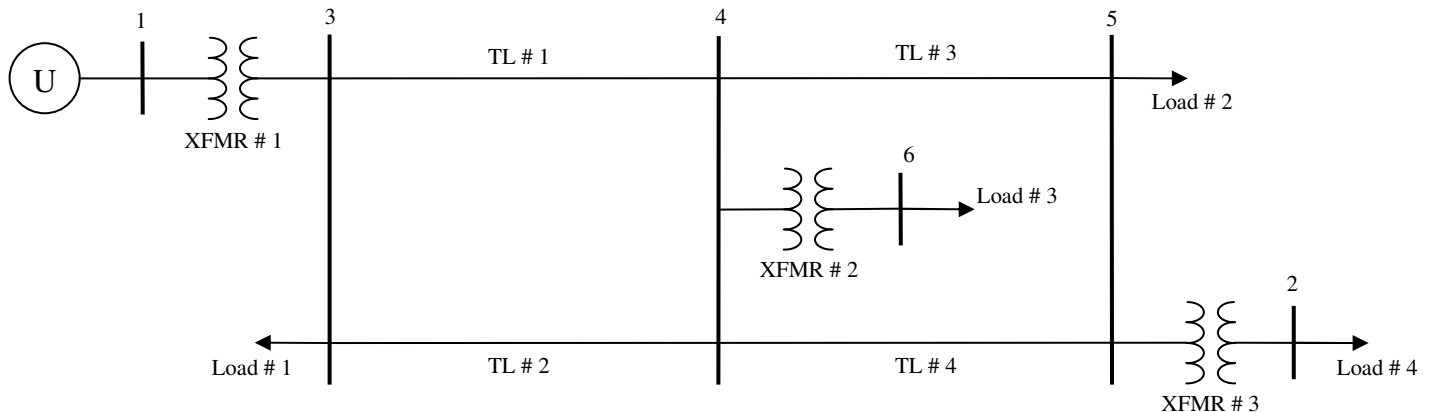
Problem #11

A 3 ϕ power system for a small manufacturing facility is supplied by a 485V tie from the electric utility company. Inside the manufacturing facility, there are three power distribution panels to which all of the building's loads are connected. The first distribution panel is connected to the electric utility company tie through a 500ft power cable, and the second and third distribution panels are connected to the first distribution panel through separate 350ft and 475ft power cables.

Draw and completely label all four of the different types of power system diagrams (block diagram, one line diagram, wiring diagram, and equivalent circuit) for this 1 ϕ power system. Use the series impedance model for the power cables in the equivalent circuit.

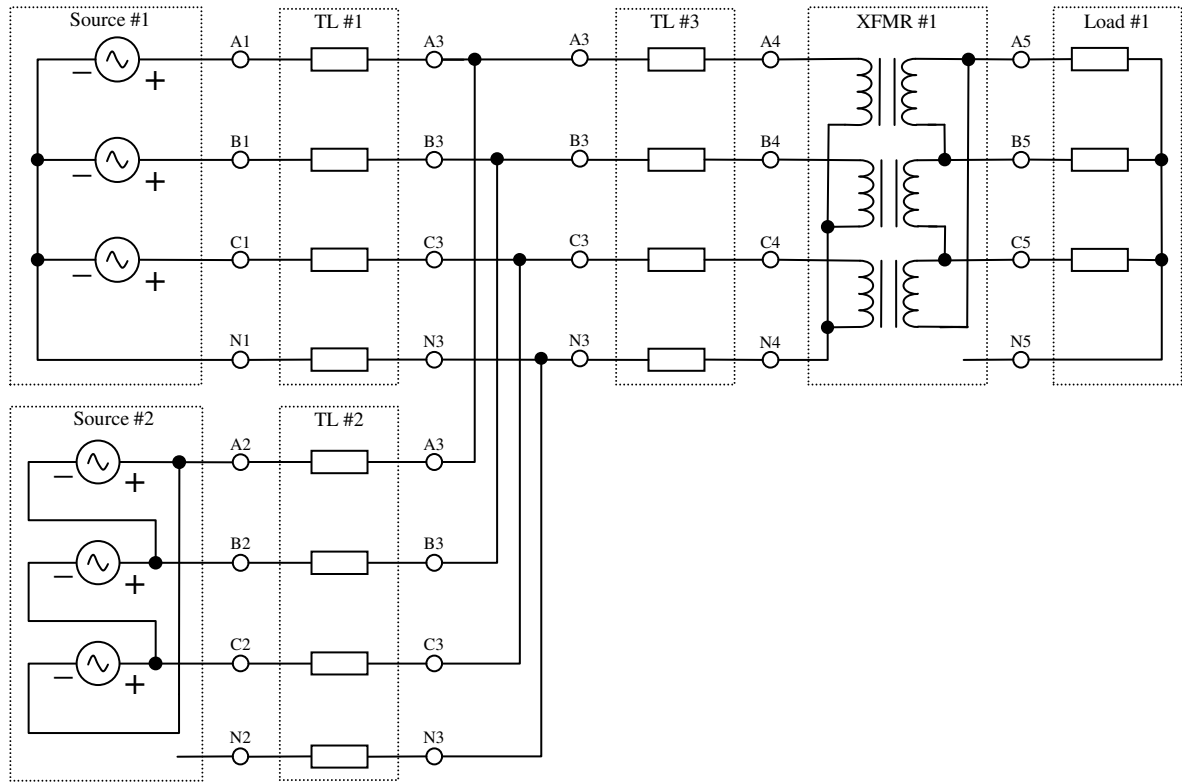
Problem #12

The one line diagram for a 3 ϕ power system is shown below. Draw the **equivalent circuit** for this 3 ϕ power system. In the equivalent circuit, Use the pi model for transmission lines TL#1 and TL#4, and use the series impedance model for transmission lines TL#2 and TL#3.



Problem #13

The wiring diagram for a 3 ϕ power system is shown below. Draw a **one line diagram** and the **equivalent circuit** for this 3 ϕ power system. In the equivalent circuit, Use the pi model for transmission lines TL#1 and TL#2, and use the series impedance model for transmission line TL#3.



Problem #14

The wiring diagram for a 1 ϕ power system is shown below. Draw a **one line diagram** and the **equivalent circuit** for this 1 ϕ power system. In the equivalent circuit, Use the pi model for transmission line TL#1, and use the series impedance model for the other transmission lines.

